

Template Exam Solutions Document

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Evaluated criteria

C1 Example criterion description.

C2 Example criterion description.

C3 Example criterion description.

1 Example Uniform Investment Plan [C1-3 C2-2]

A variable V follows a uniform model on $[8, 16]$. Use the graph and density model to answer the tasks.

Questions/Tasks.

C1 Write the density function and support. [3]

C2 Compute $P(10 \leq V \leq 14)$ and justify with area. [2]

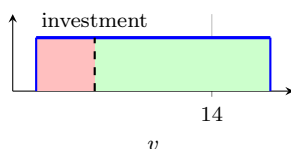
C1 - Write the density function and support. [3]

$$f_V(v) = \begin{cases} \frac{1}{16-8} = \frac{1}{8}, & 8 \leq v \leq 16, \\ 0, & \text{otherwise.} \end{cases}$$

$$8 \leq v \leq 16 \quad \longrightarrow \quad f_V(v) = \frac{1}{8} \quad \longrightarrow \quad \int_8^{16} f_V(v) dv = 1$$



C2 - Compute $P(10 \leq V \leq 14)$ and justify with area. [2]



$$P(10 \leq V \leq 14) = \int_{10}^{14} \frac{1}{8} dv = \frac{1}{8}(14 - 10) = \frac{4}{8} = 0.5$$

The shaded area represents half of the support interval.



2 Example Linear Density Model

[C2-5]

Suppose X has density $f(x) = kx$ for $0 \leq x \leq 2$.

Questions/Tasks.

C2 Find the value of k .

[3]

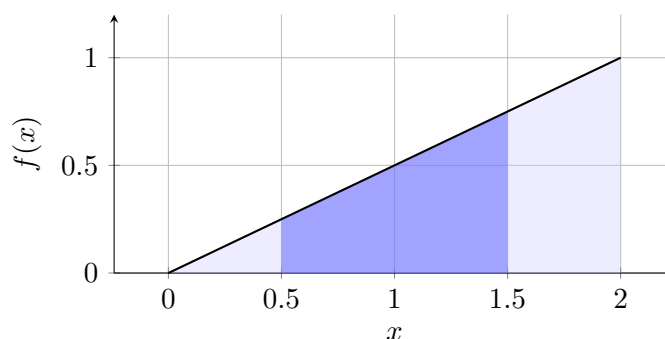
C2 Compute $P(0.5 \leq X \leq 1.5)$.

[2]

C2 - Find the value of k

[3]

$$1 = \int_0^2 kx \, dx = k \left[\frac{x^2}{2} \right]_0^2 = 2k \Rightarrow k = \frac{1}{2}.$$



C2 - Compute $P(0.5 \leq X \leq 1.5)$

[2]

$$P(0.5 \leq X \leq 1.5) = \int_{0.5}^{1.5} \frac{1}{2}x \, dx = \left[\frac{x^2}{4} \right]_{0.5}^{1.5} = \frac{1.5^2 - 0.5^2}{4} = 0.5$$



3 Example Table and Probability Summary

[C3-3]

A summary table is shown for two intervals of a piecewise density.

Questions/Tasks.

C3 Complete the probability summary table.

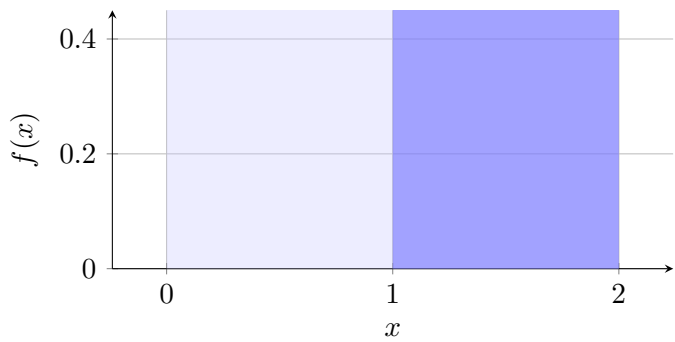
[1]

C3 State one consistency check.

[2]

C3 - Complete the probability summary table

[1]



Interval	Expression	Result
$[0, 1]$	$\int_0^1 0.4 \, dx$	0.4
$[1, 2]$	$\int_1^2 0.6 \, dx$	0.6
Total	$0.4 + 0.6$	1



C3 - State one consistency check

[2]

A valid check is that all interval probabilities are nonnegative and add to 1.

